Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 31: Approved Solutions

The Multiplication and Addition Principles

Covers Chartrand & Zhang, Section 8.1

Status of this set

Ungraded practice. Solutions released with it. Show the product or sum, not just the number — the setup is where counting goes wrong.

Part I: How to Practice This Material

Which principle?

"And then" ⇒ multiply. **"Or," non-overlapping** ⇒ add.
Before multiplying, ask: *is the number of choices at each stage the same no matter what was chosen earlier?* If not, split into cases and add.

Three mistakes to expect

1. **Adding when you should multiply**, or the reverse.
2. **Forgetting a restriction** — a leading digit cannot be 0.
3. **Multiplying when a stage's count varies.** That calls for cases.

Part II: Practice Problems

Section A — Direct counts

1. Count each, showing the product or sum.

(a) Passwords of exactly 6 characters, each a lowercase letter or a digit.

(b) Three-digit numbers whose digits are all odd.

(c) Bit strings of length 10 that begin with 1 and end with 0.

Approved solution

(a) 36 choices per character, 6 characters: $36^6 = 2{,}176{,}782{,}336$.
(b) Each digit from $\{1, 3, 5, 7, 9\}$: $5^3 = 125$. (No leading-zero issue, since 0 is not odd.)

(c) First and last bits fixed, 8 free: $2^8 = 256$.

2. How many four-digit numbers have no repeated digit? How many have *at least one* repeated digit?

Approved solution

No repeat: first digit 9 choices (not 0), then 9, 8, 7: $9 \cdot 9 \cdot 8 \cdot 7 = 4536$.
At least one repeat: there are $9 \cdot 10^3 = 9000$ four-digit numbers in all, so $9000 - 4536 = 4464$.

The second part is complement counting. "At least one repeat" is messy to count directly — it could be one repeat, two, a triple — but its complement is exactly the first part. Wednesday's lecture names this technique.

Section B — Addition

3. How many bit strings have length at most 4? (Count length 1 through 4.)

Approved solution

By the Addition Principle over the disjoint lengths:

$$2 + 4 + 8 + 16 = 30.$$

Each length is counted by the Multiplication Principle; the lengths are combined by addition. This is the Lecture 31 pattern: multiply within, add across.

Section C — When stages interfere

4. How many three-digit numbers are odd and have all distinct digits?

Approved solution

Choose the last digit first, since it has the most restrictive condition: 5 choices $\{1, 3, 5, 7, 9\}$. None of these is 0, so the first digit is unaffected by a zero.
First digit: cannot be 0 and cannot equal the last digit, so $10 - 2 = 8$ choices. Middle digit: any digit except the two used, so 8 choices.
Total: $5 \cdot 8 \cdot 8 = 320$.

Compare Lecture 31's even-digit count, which needed cases because 0 was one of the even options. Here no odd digit is 0, so the stages do not interfere and one product works. Knowing when you need cases is the real skill.

5. Explain why the count of three-digit *even* numbers with distinct digits cannot be computed as a single product, and then compute it.

Approved solution

The last digit must be even: $0, 2, 4, 6$, or 8. If it is 0, the first digit has 9 choices; if it is one of the other four, the first digit has only 8, since it cannot be 0 or equal the last digit. The number of choices for the first digit *depends* on the last, so no single product applies.
Case 1: last digit 0**.** First 9, middle 8: 72.

Case 2: last digit in $\{2, 4, 6, 8\}$. Last 4, first 8, middle 8: 256.
Total by the Addition Principle: $72 + 256 = 328$.

Section D — Writing

6. In four to six complete sentences, explain how the Multiplication Principle relates to the Cartesian product from Lecture 11, and why the count of subsets of an n-element set is 2^n.

Approved solution

Model answer. A procedure with two stages, where the first has n_1 outcomes and the second has n_2, produces an ordered pair of outcomes, so the set of all results is a Cartesian product and its size is $n_1 n_2$. The Multiplication Principle is therefore $|A \times B| = |A| \cdot |B|$ extended to any number of stages. To count subsets of an n-element set, build a subset by deciding, for each element in turn, whether it is in or out: n stages, each with 2 choices. The Multiplication Principle gives $2 \cdot 2 \cdots 2 = 2^n$, which is Theorem 2.17. This is the same argument as the four-character binary password from the first day of the course.

Full credit requires the ordered-pair connection and the in-or-out argument.

Instructor notes

- **Problems 4 and 5 are the pair to discuss.** Odd distinct digits need no cases; even distinct digits do. The difference is whether 0 is among the options.
- **Problem 2's second part** previews complement counting before Wednesday names it.
- **Problem 6** ties the principle back to Lecture 11 and the Day 1 password.
- Reserved for Homework 12: Section 8.1 even-numbered exercises. Odd ones have printed answers and are self-check only.